

SQL Business Analysis Project

Prepared By:

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Tools Used:

- MySQL
- MySQL Workbench
- SQL
- Superstore Dataset

Project Objective:

The objective of this project is to analyze retail sales data using SQL queries to identify sales trends, customer behaviour, product performance, and regional business insights. The project demonstrates the use of SQL concepts such as filtering, aggregation, grouping, sorting, and joins for business analysis.

SQL Concepts Used:

- Select
- Where
- Group By
- Order By
- Count
- Having
- Inner Join
- Aggregate Functions

1. Business Question:

What are the sales values across different product categories?

SQL Query:

```
SELECT category,sales FROM orders;
```

Output:

```
6 • select category,sales from orders;
```

	category	sales
▶	Furniture	261.96
	Furniture	731.94
	Office Supplies	14.62
	Furniture	957.5775
	Office Supplies	22.368
	Furniture	48.86
	Office Supplies	7.28
	Technology	907.152
	Office Supplies	18.504
	Office Supplies	114.9
	Furniture	1706.184
	Technology	911.424
	Office Supplies	15.552
	Office Supplies	407.976
	Office Supplies	68.81

orders 4 ×

Insight:

The dataset contains sales transactions across multiple product categories, providing a foundation for category-wise business analysis and performance evaluation.

2. Business Question:

How do technology products perform in terms of sales transactions?

SQL Query:

```
SELECT * FROM orders WHERE category='Technology';
```

Output:

7 • `select * from orders where category='Technology';`

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson
41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Houston
42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamias	Corporate	United States	Naperville
45	CA-2016-118255	3/11/2016	3/13/2016	First Class	ON-18715	Odella Nelson	Corporate	United States	Eagan
48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover
49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City
60	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles

orders 5 x Read On

Insight:

Technology products show strong sales performance across multiple customer segments.

3. Business Question:

Which category generates the highest total sales?

SQL Query:

```
SELECT category,sum(sales) AS Total_Sales FROM orders GROUP BY category ORDER BY Total_Sales DESC;
```

Output:

9 • `select category,sum(Sales) as Total_Sales from orders group by category order by Total_Sales desc;`

category	Total_Sales
Technology	835900.0669999964
Furniture	733046.8612999996
Office Supplies	703502.9280000031

Insight:

Technology category generated the highest total sales among all product categories, indicating strong customer demand and higher revenue contribution.

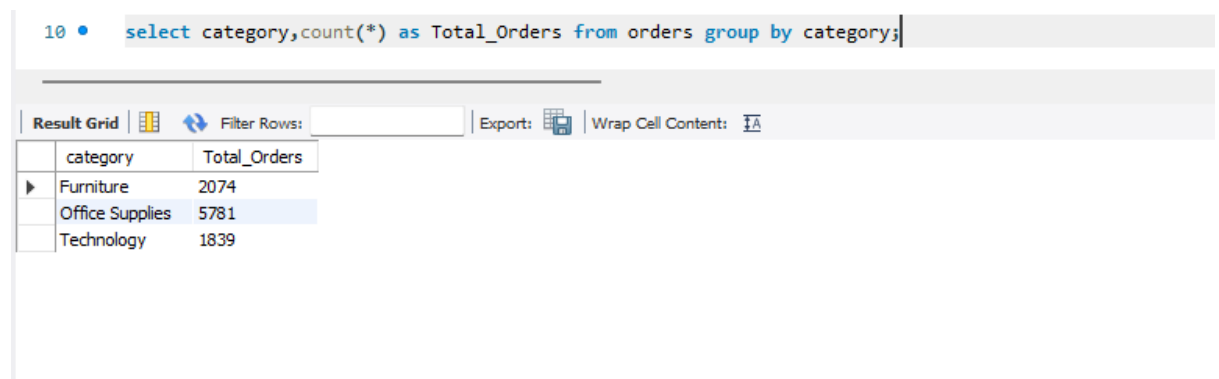
4. Business Question:

Which product category has the highest number of orders?

SQL Query:

```
SELECT category,COUNT(*) AS Total_Orders FROM orders GROUP BY category;
```

Output:



The screenshot shows a SQL query editor with the following query: `select category,count(*) as Total_Orders from orders group by category;`. Below the query, there is a toolbar with options like 'Result Grid', 'Filter Rows', 'Export', and 'Wrap Cell Content'. The results are displayed in a table with two columns: 'category' and 'Total_Orders'.

category	Total_Orders
Furniture	2074
Office Supplies	5781
Technology	1839

Insight:

Office Supplies category contains the highest number of orders, indicating frequent customer purchases.

5. Business Question:

Which regions generate higher average profits for the business greater than 30?

SQL Query:

```
SELECT region,AVG(profit) AS Avg_Profit FROM orders GROUP BY region HAVING Avg_Profit>30;
```

Output:

```
12 • select region,avg(profit) as Avg_Profit from orders group by region having Avg_Profit>30;
```

region	Avg_Profit
West	34.211406744111024
East	32.89985947024682

Insight:

West and East regions show higher average profits greater than 30 compared to other regions.

6. Business Question:

How can customer information be combined with sales data using SQL joins?

SQL Query:

```
SELECT customers.'Customer Name',orders.Sales FROM  
customers INNER JOIN orders ON customers.'Customer  
ID'=orders.'Customer ID';
```

Output:

```
41 • select customers.'Customer Name',orders.Sales from customers inner join orders on customers.'Customer ID'=orders.'Cust
```

Customer Name	Sales
Claire Gute	261.96
Claire Gute	261.96
Claire Gute	731.94
Claire Gute	731.94
Darrin Van Huff	14.62
Darrin Van Huff	14.62
Darrin Van Huff	14.62
Sean O'Donnell	957.5775
Sean O'Donnell	957.5775
Sean O'Donnell	957.5775
Sean O'Donnell	22.368
Sean O'Donnell	22.368
Sean O'Donnell	22.368
Brosina Hoffman	48.86
Brosina Hoffman	48.86
Brosina Hoffman	48.86
Brosina Hoffman	48.86
Brosina Hoffman	7.28

Result 22 x

Insight:

Certain customers contribute repeatedly to total sales through multiple purchases.

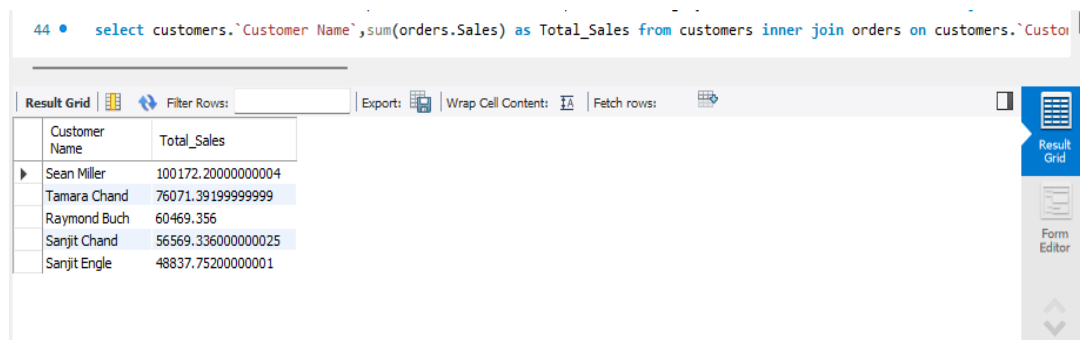
7. Business Question:

Who are the top customers contributing to overall sales revenue?

SQL Query:

```
SELECT customers.'Customer Name',SUM(orders.Sales) AS  
Total_Sales FROM customers INNER JOIN orders ON  
customers.'Customer ID'=orders.'Customer ID' GROUP BY  
customers.'Customer Name' ORDER BY Total_Sales DESC LIMIT 5;
```

Output:



Customer Name	Total_Sales
Sean Miller	100172.20000000004
Tamara Chand	76071.39199999999
Raymond Buch	60469.356
Sanjit Chand	56569.3360000000025
Sanjit Engle	48837.75200000001

Insight:

A small group of customers contributes significantly to overall sales revenue.

8. Business Question:

How can customer, product, and sales data be combined to analyze purchasing behavior across different product categories?

SQL Query:

```
SELECT customers.'Customer Name',products.'Product  
Name',products.Category,orders.Sales FROM orders INNER JOIN  
customers ON customers.'Customer ID'=orders.'Customer ID' INNER  
JOIN products ON products.'Product ID'=orders.'Product ID';
```

Output:

```
43 • select customers.`Customer Name`,products.`Product Name`,products.Category,orders.Sales from orders inner join customer
```

Customer Name	Product Name	Category	Sales
Claire Gute	Hon Deluxe Fabric Upholstered Stacking Chairs,...	Furniture	731.94
Claire Gute	Bush Somerset Collection Bookcase	Furniture	261.96
Darrin Van Huff	Self-Adhesive Address Labels for Typewriters b...	Office Supplies	14.62
Sean O'Donnell	Eldon Fold 'N Roll Cart System	Office Supplies	22.368
Sean O'Donnell	Bretford CR4500 Series Slim Rectangular Table	Furniture	957.5775
Brosina Hoffman	Konftel 250 Conference phone - Charcoal black	Technology	911.424
Brosina Hoffman	Chromcraft Rectangular Conference Tables	Furniture	1706.184
Brosina Hoffman	Belkin F5C206VTEL 6 Outlet Surge	Office Supplies	114.9
Brosina Hoffman	DXL Angle-View Binders with Locking Rings by S...	Office Supplies	18.504
Brosina Hoffman	Mitel 5320 IP Phone VoIP phone	Technology	907.152
Brosina Hoffman	Newell 322	Office Supplies	7.28
Brosina Hoffman	Eldon Expressions Wood and Plastic Desk Acces...	Furniture	48.86
Andrew Allen	DAX Wood Document Frame	Furniture	27.46
Andrew Allen	Eldon Executive Woodline II Desk Accessories, ...	Furniture	27.46
Andrew Allen	Xerox 1967	Office Supplies	15.552
Irene Maddox	Translucent Push Pins by OIC	Office Supplies	3.96
Irene Maddox	Tops White Computer Printout Paper	Office Supplies	146.73
Irene Maddox	Fellowes Bankers Box Staxsteel Drawer File/S...	Office Supplies	324.9

Result 24 × Read Only

Insight:

Customers purchase products from multiple categories, and combining customer, product, and sales data provides deeper insights into customer buying patterns and product performance.

Key Insights:

- Technology category generated the highest total sales.
- Office Supplies category had the highest number of orders.
- West and East regions showed stronger profitability.
- A small group of customers contributed significantly to total revenue.
- SQL joins helped combine customer, product, and sales data for deeper business analysis.

Conclusion:

This project demonstrates the practical application of SQL for business data analysis using retail sales data. SQL concepts such as filtering, aggregation, grouping, sorting, and joins were used to generate meaningful business insights and improve analytical problem-solving skills.